

PLYMOUTH, MA, USA

WMO: 725064

Lat: 41.910N

Lon: 70.729W

Elev: 149

StdP: 14.62

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 54769

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.5	10.2	-9.2	3.4	10.3	-4.6	4.3	14.2	27.8	28.9	25.3	33.4	7.0	290	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	89.0	73.1	85.7	71.8	82.6	70.0	76.0	84.5	74.5	81.5	73.1	79.2	10.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.2	124.2	79.3	72.3	120.4	78.3	71.3	116.2	77.2	39.4	84.8	38.0	81.8	36.7	79.2	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	20.0	17.9	-0.5	94.4	5.6	3.3	-4.6	96.8	-7.9	98.7	-11.0	100.5	-15.1	102.9
			-2.0	78.4	5.9	1.7	-6.3	79.7	-9.7	80.7	-13.0	81.6	-17.2	82.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.3	29.1	30.8	36.6	46.5	56.4	65.4	72.0	70.7	63.6	52.8	43.2	34.6		
	DBStd	16.90	10.26	9.52	8.68	7.50	7.50	6.93	5.69	5.73	6.90	7.79	8.54	9.10		
	HDD50	2575	650	540	426	156	21	0	0	0	1	58	238	485		
	HDD65	5988	1114	958	882	558	285	79	9	15	107	385	653	943		
	CDD50	2668	2	2	9	50	220	463	683	642	409	146	35	7		
	CDD65	603	0	0	0	1	19	91	227	192	65	8	0	0		
	CDH74	4597	0	0	2	31	211	741	1818	1389	371	34	0	0		
	CDH80	1329	0	0	0	8	56	217	590	383	72	3	0	0		
(14) Wind	WSAvg	7.8	8.8	8.8	9.3	8.6	7.7	7.0	6.6	6.1	6.5	7.5	7.9	8.2		
(15) Precipitation	PrecAvg	50.8	4.6	4.3	4.6	4.4	3.7	3.4	3.2	4.0	4.4	4.3	4.7	4.9		
	PrecMax	70.2	11.6	10.2	13.6	11.8	9.3	10.8	8.0	9.4	13.3	12.0	10.0	11.3		
	PrecMin	28.3	1.3	1.3	0.2	1.1	0.4	0.5	0.9	0.2	0.9	0.7	1.1	1.2		
	PrecStd	9.3	2.5	2.0	2.8	2.2	1.9	2.5	1.6	2.1	2.7	2.4	2.4	2.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.0	58.6	69.8	78.9	86.6	90.7	93.2	92.3	86.5	78.7	68.2	62.7		
		MCWB	56.3	53.9	56.7	63.0	68.1	73.2	75.0	76.0	72.7	66.2	61.8	58.4		
	2%	DB	54.2	53.9	59.3	68.9	80.0	85.7	89.7	87.2	81.5	72.9	64.1	57.5		
		MCWB	51.2	49.1	49.7	56.2	65.0	71.2	73.3	73.2	69.3	65.1	59.4	54.6		
	5%	DB	49.1	49.6	54.2	63.7	74.5	81.8	86.2	84.0	78.7	69.6	61.1	54.1		
		MCWB	45.7	45.5	48.1	53.2	62.1	69.0	72.4	71.3	68.3	63.1	56.3	51.3		
	10%	DB	44.5	45.2	50.2	59.3	70.0	78.4	82.9	81.4	74.8	65.8	57.3	49.6		
		MCWB	41.1	40.2	44.6	50.9	59.6	66.3	70.6	70.1	67.4	60.2	52.8	46.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.8	55.7	58.1	63.8	71.0	76.0	78.4	77.8	74.9	70.2	64.2	59.9		
		MCDB	58.4	57.8	64.6	72.1	82.7	87.2	90.1	88.9	82.0	74.5	66.4	62.0		
	2%	WB	52.1	50.5	53.4	59.5	67.3	73.3	76.2	75.9	72.8	67.1	61.0	55.3		
		MCDB	53.6	52.5	57.2	66.1	76.2	82.5	85.2	83.4	77.8	71.1	63.5	57.5		
	5%	WB	46.2	45.8	49.6	56.2	64.4	71.1	74.5	74.5	71.1	64.3	57.5	51.5		
		MCDB	48.3	48.4	53.3	60.7	71.5	78.2	82.0	80.4	75.5	67.8	60.3	53.2		
	10%	WB	41.2	41.0	44.9	52.9	61.6	69.1	73.1	73.0	69.4	61.6	54.0	46.6		
		MCDB	44.3	44.7	48.6	57.6	67.7	75.0	79.5	78.0	73.8	65.4	56.8	48.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.9	17.1	16.9	18.6	18.9	18.5	18.3	17.9	18.2	18.5	17.2	16.0		
		MCDBR	20.6	21.4	22.8	26.3	26.0	23.7	21.5	20.5	20.9	21.1	18.9	19.2		
	5% WB	MCWBR	19.2	18.0	16.2	15.1	13.3	11.5	8.9	9.0	11.4	14.0	15.2	18.2		
		MCWBR	19.9	20.3	20.9	21.1	22.7	20.2	17.9	16.8	16.4	18.0	17.2	18.7		
(40) Clear-Sky Solar Irradiance	taub		0.301	0.308	0.331	0.369	0.405	0.456	0.478	0.433	0.372	0.339	0.314	0.301		
	taud		2.513	2.500	2.481	2.396	2.337	2.232	2.177	2.291	2.442	2.538	2.570	2.565		
	Ebn at Noon		271	287	292	286	277	262	254	263	273	272	265	262		
	Edh at Noon		23	27	31	36	39	44	46	40	32	26	22	20		
(44) All-Sky Solar Radiation	RadAvg		562	842	1163	1502	1682	1814	1887	1697	1372	926	625	471		
	RadStd		42	79	85	161	170	169	108	82	94	79	45	49		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+236	N/A
Regional (65 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+79

Nomenclature: See separate page